Phosphate

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Identity Sheet

Property	Value
Name	Phosphate
Other names	Inorganic phosphate, orthophosphate, Pi
Chemical formula	PO_4^{3-} (fully deprotonated); $H_2PO_4^-$ and HPO_4^{2-} at physiological pH
Molecular weight	94.97 g/mol (PO_4^{3-})
CAS number	14265-44-2 (phosphate ion)
PubChem CID	1061
Appearance	Colourless in solution

Structural formula:

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O(-)
|
(-)O-P-O(-) (tetrahedral geometry)
|
O(-)
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Phosphate is an essential mineral anion. At urinary pH (4.5-8.0), it exists primarily as a mixture of $\mathbf{H_2PO_4^-}$ (dihydrogen phosphate, pKa_2 = 6.8) and $\mathbf{HPO_4^{2-}}$ (monohydrogen phosphate). This buffering pair constitutes the major **titratable acid** system in urine, critical for renal acid excretion.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
Pyrophosphate (PPi)	$P_2O_7^{4-}$	173.94	Condensation dimer; inhibitor of calcification; different metabolism
Organic phosphates (ATP, etc.)	Variable	Variable	Phosphorylated organic molecules; not measured by inorganic Pi assays
Phosphonate	RPO_3^{2-}	Variable	C-P bond (not C-O-P); synthetic compounds

Medical Information

Origin

Endogenous Total body phosphorus: ~700 g in adults (85% in bone as hydroxyapatite, 14% in soft tissue, 1% in ECF). No endogenous synthesis – phosphorus is an element. Released from bone remodelling and intracellular stores.

Exogenous Dietary phosphate from dairy, meat, fish, eggs, legumes, nuts, cola drinks. Typical intake: 800-1500 mg/day. Intestinal absorption: 60-70% (enhanced by vitamin D, inhibited by calcium and aluminium).

Primary & Secondary Biological Roles

Primary role: - **Bone mineralisation:** Hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$) is the structural mineral of bones and teeth. - **Energy metabolism:** ATP, GTP, creatine phosphate – phosphate is the energy currency bond.

Secondary roles: - **Urinary buffer:** $H_2PO_4^-/HPO_4^{2-}$ system (pKa 6.8) is the primary titratable acid buffer in urine. - **Nucleic acids and membranes:** DNA/RNA backbone; phospholipid bilayers. - **Intracellular signalling:** Protein phosphorylation; second messengers (cAMP, IP_3). - **2,3-DPG in RBCs:** Regulates oxygen release from haemoglobin.

Catabolism and Elimination Pathway

- Phosphate freely filtered at glomerulus (~6,500 mg/day).
- **80-90% reabsorbed** in proximal tubule via NaPi-IIa and NaPi-IIc transporters (regulated by PTH and FGF23, which reduce reabsorption).
- Remaining 10-20% excreted in urine (major excretion route).
- Minor GI losses. No significant sweat excretion.
- **PTH** increases urinary phosphate (phosphaturic). **FGF23** also increases phosphate excretion. **Vitamin D** increases intestinal absorption and may increase urinary excretion secondarily.

Expression in Humans

Normal Levels

Compartment	Reference Range
Serum phosphate	2.5-4.5 mg/dL (0.81-1.45 mmol/L) in adults
Urinary phosphate (24-h)	400-1300 mg/day (13-42 mmol/day)
TmP/GFR (tubular max reabsorption)	2.5-4.2 mg/dL
FEPi (fractional excretion)	5-20%

Factors Influencing Levels **Increased urinary phosphate:** - Hyperparathyroidism (PTH promotes phosphaturia) - FGF23-producing tumours (tumour-induced osteomalacia) - Fanconi syndrome (proximal tubular dysfunction) - High phosphate diet - Metabolic acidosis (shifts phosphate from cells to ECF) - Vitamin D excess (increased absorption, overflow)

Decreased urinary phosphate: - Hypoparathyroidism (increased reabsorption) - Vitamin D deficiency - Low phosphate diet - Phosphate binders (aluminium, calcium, sevelamer) - Growth (anabolism)

Associated Pathologies

Condition	Phosphate Pattern	Key Symptoms
Primary hyperparathyroidism	High urinary PO4, low serum PO4	Hypercalcaemia, stones, bone pain
CKD (advanced)	Low urinary PO4, high serum PO4	Renal osteodystrophy, vascular calcification

Condition	Phosphate Pattern	Key Symptoms
Fanconi syndrome	High urinary PO4 + glucose + amino acids	Rickets/osteomalacia, growth failure
Tumour-induced osteomalacia	Very high FEPi, low serum PO4	Bone pain, fractures, muscle weakness
Calcium phosphate nephrolithiasis	Elevated urinary PO4 + Ca	Kidney stones (brushite, apatite)

Presence in Urine

Should it be normally present? Yes – urinary phosphate excretion is the main regulatory mechanism for phosphate balance.

Normal urinary levels: 400-1300 mg/day (13-42 mmol/day).

Form in urine: Mixture of $\mathbf{H_2PO_4^-}$ and $\mathbf{HPO_4^{2-}}$ (ratio depends on urine pH; at pH 6.8, equal amounts). Calcium phosphate crystals (amorphous calcium phosphate, brushite $CaHPO_4$) can form in alkaline urine.

Pathological significance:

Urinary Phosphate	Possible Causes	Prevalence
High (>1300 mg/day)	Hyperparathyroidism, Fanconi, high diet	HPT: ~1:1000
Low (<400 mg/day)	Vitamin D deficiency, low intake, CKD	Common

Solubility: Phosphate salts have variable solubility. Calcium phosphate precipitates in **alkaline urine** (pH >7), contributing to nephrolithiasis. Struvite ($MgNH_4PO_4$) crystals form in infected alkaline urine.

Detection in Urine

Available Clinical Assays

1. **Phosphomolybdate spectrophotometric assay (gold standard):**
   - **Principle:** Phosphate reacts with ammonium molybdate in acid -> phosphomolybdate complex, reduced to molybdenum blue.
   - **Detection:** Absorbance at **340 nm** (UV, unreduced complex) or **660-700 nm** (molybdenum blue, reduced).
   - **LOD:** ~0.1 mg/dL (~0.03 mmol/L).
   - **Advantages:** Standard on automated platforms; specific for inorganic Pi.
2. **Enzymatic assay (purine nucleoside phosphorylase):**
   - **Principle:** Pi + inosine (PNP) -> hypoxanthine (XOD) -> uric acid + H_2O_2 -> colour (Trinder).
   - **Detection:** 546 nm.
   - **LOD:** ~0.05 mg/dL.
   - **Advantages:** More specific, fewer interferences.

Optimal Urine Type for Measurement

- **24-hour urine** for total phosphate excretion (stone risk assessment, metabolic workup).
- Spot urine for TmP/GFR calculation (with simultaneous serum phosphate and creatinine).
- Acidify urine with HCl during collection to prevent phosphate crystal precipitation.
- Refrigerate.

Actual Gold Standard

Phosphomolybdate UV method (340 nm, unreduced) on automated platforms (Roche, Siemens). CV: 2-4%.

Interferences in Measurement

Interference	Effect	Mechanism
Haemolysis	Positive bias	Intracellular phosphate release from RBCs
Lipaemia	Positive bias	Light scattering
Bilirubin	Variable	Absorbs near 340 nm
Mannitol, dextran	Positive bias	Complex with molybdate
Precipitated Ca-PO4	Negative bias	Phosphate lost to crystals if urine not acidified

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **Phosphomolybdate blue (660-700 nm):** Classical colorimetric. LOD: ~0.05 mg/dL.
- **Vanadomolybdate (yellow, 400-420 nm):** Alternative method. LOD: ~0.5 mg/dL.
- **NIR:** No useful phosphate absorption.

Fluorescence Detection

- Phosphate is non-fluorescent. Indirect approaches:
- **Fluorescent chemosensors:** Europium(III) or terbium(III) complexes quenched by phosphate. Eu: Ex 340 / Em 615 nm. LOD: ~1 uM.
- **DPA-Zn complexes:** Fluorescent turn-on sensors for pyrophosphate/phosphate. Ex 340 / Em 460 nm.

Raman Detection

Peak (cm-1)	Assignment
~940	P-O symmetric stretch (v1) – strongest
~1020	P-O asymmetric stretch (v3)
~420	O-P-O bending (v2)

- Conventional Raman detects phosphate at >10 mM. The 940 cm-1 peak is diagnostic.
- LOD: ~5-10 mM (adequate for concentrated urine).
- Excitation: 532 or 785 nm.
- No SERS studies specifically for urinary phosphate.

FTIR Detection

Band (cm-1)	Assignment	Notes
~1080	PO asymmetric stretch (v3)	Overlaps with many biological molecules
~940	PO symmetric stretch (v1)	More specific

Band (cm-1)	Assignment	Notes
~560	PO bending (v4)	–

- ATR-FTIR: LOD ~5-10 mM in aqueous solution. Adequate for urine.

Voltammetry Detection

- Phosphate is electrochemically inert in the normal potential window.
- **Indirect electrochemical:** Phosphate precipitates with molybdate on electrode surfaces; the molybdate reduction current is proportional to PO_4 concentration.
  - DPV of phosphomolybdate: LOD ~0.1 uM.
  - Oxidation peak at +0.2-0.4 V vs Ag/AgCl.
- **Cobalt electrode method:** Co^{2+} forms insoluble $Co_3(PO_4)_2$; stripping voltammetry detects remaining Co^{2+}. LOD: ~0.5 uM.

Other Detection Technologies

1. **Ion chromatography:** Anion exchange + conductivity. LOD: ~0.01 mmol/L. Multi-anion analysis.
2. **ICP-OES:** Phosphorus emission at 213.6 nm. LOD: ~0.001 mg/dL.
3. **Malachite green assay:** Phosphomolybdate + malachite green -> green complex (620 nm). LOD: ~0.05 uM. Very sensitive.

Summary Table: Detection Methods Comparison

Method	LOD	Key Parameter	Sample Prep	Strengths
Phosphomolybdate UV	~0.1 mg/dL	340 nm	None	Standard, automated
Molybdenum blue	~0.05 mg/dL	660-700 nm	Reagent	Sensitive
Enzymatic	~0.05 mg/dL	546 nm	None	Specific
Raman	~5 mM	940 cm-1	None	Reagent-free
FTIR	~5 mM	1080 cm-1	None	Non-destructive
DPV (molybdate)	~0.1 uM	+0.2-0.4 V	Molybdate addition	Ultra-sensitive
Eu fluorescence	~1 uM	Ex 340/Em 615 nm	Probe	Sensitive
ICP-OES	~0.001 mg/dL	213.6 nm	Dilution	Ultimate

References

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